

Deployment Guide: Next.js WGPA Calculator

This guide provides step-by-step instructions on how to deploy your Next.js GPA/WGPA Calculator application for free using two popular platforms: **Vercel** and **Microsoft Azure**.

Phase 0: Preparation (Required for both methods)

Before deploying to either platform, your code needs to be hosted in a Git repository (GitHub is highly recommended as both platforms integrate seamlessly with it).

1. Initialize a git repository in your project folder: `git init`
2. Commit your code: `git add .` and `git commit -m "Initial commit"`
3. Create a new repository on [GitHub](#).
4. Push your local code to the GitHub repository following the instructions provided by GitHub.

Method 1: Deploying to Vercel (Recommended)

Vercel is the company behind Next.js. Deploying here is incredibly fast, fully optimized, and completely free for personal/hobby projects.

Prerequisites

- A [Vercel account](#) (Sign up using your GitHub account for the easiest integration).

Step-by-Step Deployment

1. **Log in to Vercel:** Go to your Vercel dashboard.
2. **Add New Project:** Click the "Add New..." button and select "Project".
3. **Import Git Repository:** * Vercel will ask to connect to your GitHub account.
 - Once connected, you will see a list of your repositories.
 - Find your GPA Calculator repository and click "Import".
4. **Configure Project:**
 - **Project Name:** Leave as default or rename it.
 - **Framework Preset:** Vercel will automatically detect **Next.js**. Leave it as is.
 - **Root Directory:** If your code is in the root of the repo, leave this empty.
 - **Environment Variables:** Since this is a simple client/static app, you likely won't need any environment variables right now.
5. **Deploy:** Click the "Deploy" button.
6. **Wait for Build:** Vercel will now build your application. This usually takes 1-2 minutes.
7. **Success!** Once finished, Vercel will provide you with a live, free URL (e.g., `your-app-name.vercel.app`). Any future pushes to your GitHub `main` branch will automatically trigger a new deployment.

Method 2: Deploying to Azure Static Web Apps (Free Tier)

Azure Static Web Apps natively supports Next.js (including server-side rendering features) and offers a generous free tier perfect for this type of application.

Prerequisites

- A [Microsoft Azure Account](#). You can start with a free account.

Step-by-Step Deployment

1. **Log in to Azure Portal:** Go to portal.azure.com.
2. **Create a Resource:** Click "Create a resource" in the top left menu.
3. **Search for Static Web App:** Type "Static Web App" into the search bar and select it, then click "Create".
4. **Basic Configuration:**
 - **Subscription:** Select your Free Trial or Azure subscription.
 - **Resource Group:** Click "Create new" and name it (e.g., `gpa-calculator-rg`).
 - **Name:** Give your app a name (e.g., `wgpa-calculator-s1`).
 - **Plan type:** Select **Free** (For hobby or personal projects).
 - **Region:** Select the region closest to your target users (e.g., East Asia or Central India).
5. **Deployment Details:**
 - **Source:** Select **GitHub**.
 - Click "Sign in with GitHub" and authorize Azure to access your repositories.
 - Select your **Organization**, **Repository**, and the **Branch** (usually `main` or `master`) where your code lives.
6. **Build Details:**
 - **Build Presets:** Select **Next.js**.
 - **App location:** `/` (Assuming your `package.json` is in the root folder).
 - **Api location:** Leave blank (Unless you create custom backend APIs later).
 - **Output location:** Leave blank (Azure automatically handles this for Next.js).
7. **Review and Create:**
 - Click "Review + create" at the bottom, then click "Create".
8. **Wait for GitHub Actions:** * Azure does not build the app directly. Instead, it creates a GitHub Actions workflow file in your repository.
 - Go back to your GitHub repository and click on the "Actions" tab. You will see a build process running.
9. **Success!** Once the GitHub Action completes successfully, go back to the Azure Portal. Go to your new Static Web App resource, and you will see your live URL (usually a randomly generated string like `yellow-water-1234.azurestaticapps.net`).

Post-Deployment: Custom Domains

Both Vercel and Azure allow you to add your own custom domains (e.g., `mygpacalc.com`) for free.

- **In Vercel:** Go to Project Settings > Domains.
- **In Azure:** Go to your Static Web App resource > Custom domains > Add.